

Start-up Plan

Thesis: Passivity-Based Control with Physics-Informed Structured Learning for Robot Manipulators

0. Diagnosis

The proposal is written at the level of the final thesis. You have nonlinear control and introduction to robotics, and you have not yet met three of the four bodies of literature the proposal stands on.

Body of work	Owner	Your current state
Passivity-based control of Euler–Lagrange systems, Slotine–Li	Ortega, Slotine	not seen
Neural compensation with Lyapunov learning laws	Dr. Yu’s line	not seen
Integral sliding manifolds, super-twisting, strict Lyapunov functions	Dr. Castaños’s line	not seen
Structure-preserving networks (DeLaN)	Lutter, Greydanus	not seen

You cannot start by reading the proposal’s 30 references. Five of them carry the argument and are listed as Group 1 in the appendix; read those completely and leave the rest until the phase that needs them. You will rebuild the proposal from the bottom, one mechanism at a time, in a simulator you wrote yourself. Everything below is organized around that.

1. Six governing rules

1. **One sandbox for the whole term.** A planar 2-DOF manipulator, derived by hand, simulated by you. Every phase adds one block to the same code. No new platform, no toolbox black box.
2. **Nothing is read that is not immediately implemented.** Reading a paper without reproducing its central inequality numerically does not count as having read it.
3. **Every theorem is re-derived by hand before it is used.** Keep one physical notebook of derivations. Either advisor may ask you to reproduce any of them at the board.
4. **Every phase ends with a numerical exit test with a pass/fail number.** No exit test, no next phase.
5. **Every phase produces a 1–2 page note in L^AT_EX.** At the end of the term these notes are already Chapters 2–3 of the thesis and the state-of-the-art document.
6. **The order is: mechanism, then combination.** Do not touch the full architecture (7) until Phase 6. Assembling first is how you end up unable to say which block is responsible for what.

2. Background to fill in parallel

The proposal uses four tools that are usually not covered in a first course. Khalil, *Nonlinear Systems*, 3rd ed.:

- **Non-autonomous Lyapunov, UUB, Barbalat’s lemma** — Ch. 4 (§4.8, 4.9) and Ch. 8 (§8.4). You need UUB because the whole neural-compensation line concludes in a ball, not at the origin.

- **Passivity, $\mathcal{L}_2$ gain, output strict passivity** — Ch. 6 (§6.1–6.5). This is the language of objective 4, “prove the closed loop is output strictly passive”.
- **Filippov solutions and the equivalent control method** — operational minimum only: what a discontinuous right-hand side means, why $\text{sign}(0)$ is a set. The first chapter of Utkin’s book, or Dr. Castaños’s course notes, is enough. Do not go into differential inclusions theory.
- **Homogeneity and finite-time stability** — a two-page idea: why $\dot{V} \leq -cV^{1/2}$ gives finite time. You will need it for the super-twisting proof.

Fix the notation of the thesis before you write a line of code, and never change it: $q, e, s, \Lambda, \tilde{\Delta}, \hat{W}, \Sigma_\theta$ exactly as in the proposal.

3. The eight phases

Each phase: goal $\rightarrow$ read $\rightarrow$ derive by hand $\rightarrow$ code $\rightarrow$ exit test $\rightarrow$ deliverable.

Phase 0 — The sandbox and the three structural properties

Goal. A simulator you trust, and physical understanding of properties (2)–(5), which are the hypotheses of every proof in the thesis.

Read. `Spong, Hutchinson & Vidyasagar [24]`, Ch. 6 (Lagrangian dynamics) and Ch. 7 (properties). `Kelly, Santibáñez & Loria [7]`, Ch. 3–4. Nothing else.

Derive by hand. The full $M(q)$, $C(q, \dot{q})$, $g(q)$ of a 2-DOF planar arm with gravity, from the Lagrangian. Then derive the Christoffel form

$$c_{kj} = \sum_i \frac{1}{2} \left(\frac{\partial m_{kj}}{\partial q_i} + \frac{\partial m_{ki}}{\partial q_j} - \frac{\partial m_{ij}}{\partial q_k} \right) \dot{q}_i,$$

and prove on paper that $\dot{M} - 2C$ is skew-symmetric for this C .

Code. `robot2dof.m`: functions for $M(q)$, $C(q, \dot{q})$, $g(q)$, friction $F(\dot{q})$ (viscous + Coulomb + Stribeck), and a fixed-step RK4 integrator you write yourself. Derive the same M, C, g a second time with the Symbolic Math Toolbox (`syms`, `jacobian`) and export them with `matlabFunction`: if the symbolic result and your hand derivation disagree, you have found a mistake before it costs you a month. Add an energy monitor $H = \frac{1}{2} \dot{q}^\top M \dot{q} + U(q)$.

Exit test.

- $|x^\top (\dot{M} - 2C)x| < 10^{-10}$ for 1000 random $(q, \dot{q}, x)$.
- With $\tau = 0$, $F = 0$: relative drift of H below 10^{-6} over 10 s.
- Numerical values of $\lambda_m, \lambda_M, k_c, k_g, k_f$ tabulated over the operating set.

Deliverable. Note 1, “Model and structural properties of the test platform”. The numbers $\lambda_m, \lambda_M, k_c, k_g$ go straight into the design recipe, Section 5.8 of the proposal.

Why this matters: Section 5.1 of the proposal asks you to reproduce the Christoffel construction inside a network. If you have not built it by hand for a real arm first, DeLaN will be magic to you.

Phase 1 — Passivity and the energy argument

Goal. Understand why passivity is the object being protected, before anything threatens it.

Read. Ortega, Loria, Nicklasson & Sira-Ramírez [15], Ch. 1 and the EL chapter. Kelly et al. [7] on PD with gravity compensation. Do not read [16] (IDA-PBC / port-Hamiltonian) — it is a different branch and will cost you a great deal of time for nothing.

Derive by hand. (i) Passivity of $\tau \mapsto \dot{q}$ with storage H . (ii) Global asymptotic stability of PD + gravity compensation for regulation. (iii) The energy-shaping plus damping-injection reading of the same controller.

Code. PD+g regulation on the sandbox. Plot $V(t)$ and $\dot{V}(t)$. Verify the passivity inequality $\int_0^t \dot{q}^\top \tau d\sigma \geq H(t) - H(0)$ pointwise in time, numerically.

Exit test. The passivity inequality holds at every sample; $\dot{V} \leq 0$ at every sample; error $\rightarrow 0$ for 20 random initial conditions.

Deliverable. Note 2, “Passivity of the EL manipulator and the nominal PBC loop”.

Phase 2 — The filtered error s and the baseline to beat

Goal. Produce the baseline radius ρ_1 of eq. (10) — the first real result of the thesis.

Read. Slotine & Li [22]. Slotine & Sastry [23]. Both are short; both must be read line by line.

Derive by hand. Starting from (1) and (7) with $\hat{\Delta}_\theta = 0$, $\tau_{\text{sta}} = 0$, obtain the error dynamics $M\dot{s} = -Cs - K_D s - \Delta$. Then, with $V = \frac{1}{2}s^\top Ms$, redo the Young inequality step and obtain

$$\limsup_{t \rightarrow \infty} \|s(t)\| \leq \sqrt{\frac{\bar{\eta}_1}{\lambda_{\min}(K_D - \Lambda_1)}} =: \rho_1.$$

You must be able to say exactly where each hypothesis (2)–(5) is used.

Code. Implement (7) with the nominal terms only. Then run the detuning experiment: scale $\hat{M}$, $\hat{C}$, $\hat{g}$ by 0.9, 0.8, 0.7, 0.5 and add the friction model to the plant but not to the controller. For each detuning, sweep K_D and measure $\limsup \|s\|$.

Exit test. A plot of measured $\limsup \|s\|$ versus K_D for each detuning level, with the predicted ρ_1 overlaid. The measured curve must sit below the bound and decay like $1/\sqrt{\lambda_{\min}(K_D)}$.

Deliverable. Note 3 plus the baseline figure of the thesis. This figure is the reference against which every later claim (B1)–(B4) is made. Fix the reference trajectory, the friction parameters, the noise seed and the sampling period now, and never change them again.

Phase 3 — Neural compensation with a Lyapunov learning law

Goal. Understand that the learning law comes from $\dot{V}$, not from backpropagation, and see the ball shrink.

Read, in this order.

- Yu & Rosen [30]. Start here: it is the closest to the proposal’s Remark 7.
- Lewis, Yesildirek & Liu [10] — the canonical UUB proof with a two-layer net.
- Sanner & Slotine [20] — why a linearly parameterized regressor removes local minima.
- Yu & Li [27] — the identification-side version of the same argument.

Derive by hand. With $V_\ell = \frac{1}{2}s^\top Ms + \frac{1}{2}\text{tr}(\tilde{W}^\top K_w^{-1}\tilde{W})$, reproduce eq. (16) of the proposal, including the identity $s^\top \tilde{W}\Sigma = \text{tr}(\tilde{W}^\top s\Sigma^\top)$, and show that (17) makes the trace term vanish identically. Then answer in writing, in your own words:

- Why is only the output layer adapted?
- Why does the error converge to a ball of radius proportional to $\bar{\varepsilon}$ and not to zero?
- What exactly does σ - or e -modification add to $\dot{V}_\ell$, and what does it cost?

Code. Add an RBF or sigmoid regressor $\Sigma(x)$ with $x = [q^\top, \dot{q}^\top, \ddot{q}^\top]^\top$, adapted by (17). No optimizer and no training loop: stack $\hat{W}$ into the state vector of the same fixed-step integrator that carries the plant, so the weights move in closed loop exactly as the theory says. Apply the projection onto $\|\hat{W}\|_F \leq W_{\max}$ after each integration step.

Exit test.

- At the same K_D as in Phase 2, steady $\|s\|$ falls by at least a factor of 3.
- Turning off projection and injecting measurement noise reproduces weight drift; turning on σ -modification stops it. Show both plots.

Deliverable. Note 4, “Lyapunov-based neural compensation: the mechanism and its limit”. The closing sentence must be the proposal’s own: the ball can be shrunk by learning but must be removed by something with integral character.

Phase 4 — Sliding modes, super-twisting, strict Lyapunov functions

Goal. Understand chattering, relative degree, the gain conditions, and real sliding accuracy. From this phase on, Dr. Castaños joins the meetings.

Read, in this order.

- Levant [8] — sliding order and sliding accuracy. The $O(h^2)$ statement lives here.
- Moreno & Osorio [13] — the strict Lyapunov function; this is the one you must be able to reproduce at the board.
- Castaños & Fridman [2] — integral sliding manifolds and unmatched perturbations. Read it before your first meeting with Dr. Castaños.
- Nagesh & Edwards [14] — multivariable STA under a matrix input gain, needed for $M^{-1}(q)$.
- González, Moreno & Fridman [4] — variable-gain STA, the declared fallback.

Derive by hand. (i) Why s has relative degree one with respect to τ and why that permits a second-order algorithm with continuous output. (ii) The Moreno–Osorio quadratic form $\zeta^\top P\zeta$ with $\zeta = [s]^{1/2}, v]^\top$ and the conditions on k_1, k_2 given $\|\frac{d}{dt}\tilde{\Delta}\| \leq L$.

Code.

- First-order SMC on s : observe chattering; measure total variation of τ .
- STA (18): observe continuity; measure total variation again.
- **Gain-condition experiment:** inject an analytically known disturbance with a known Lipschitz constant L . Sweep (k_1, k_2) , find the empirical minimum gains that still give finite-time convergence, and compare with $k_2 > c_1 L, k_1 > c_2 \sqrt{L}$.
- **Sampling experiment:** sweep the sampling period h , measure the steady $|s|$, plot log–log.

Exit test.

- Total variation of τ under STA at least one order of magnitude below first-order SMC.
- Log–log slope of $|s|$ versus h equal to 2 ± 0.2 .

Deliverable. Note 5 plus the $O(k^2h^2)$ figure. This is contribution (B4) of the proposal, and you will have produced the evidence for it here, in simulation, before writing a single line of theory about it.

Phase 5 — Structured learning (DeLaN) and the offline problem

Goal. A network whose $\hat{M}_\theta$ is positive definite and whose $\dot{\hat{M}}_\theta - 2\hat{C}_\theta$ is skew-symmetric by construction, verified to machine precision.

Read. Lutter, Ritter & Peters [11], the main one. Greydanus et al. [5] and Cranmer et al. [3] as antecedents, one afternoon each. Raissi et al. [18] read only to see the contrast: penalty in the cost versus structure in the architecture. Shi et al. [21] for spectral normalization. Atkeson, An & Hollerbach [1] and Middleton & Goodwin [12] for the acceleration-free regression.

Derive by hand. (i) Why $\hat{M}_\theta = L_\theta L_\theta^\top + \epsilon I$ with L_θ lower triangular and positive diagonal gives symmetry and positive definiteness for every θ . (ii) Why $\hat{C}_\theta$ built from the Christoffel symbols of $\hat{M}_\theta$ makes $\dot{\hat{M}}_\theta - 2\hat{C}_\theta$ skew-symmetric for every θ . (iii) The filtering of (1) by $\lambda/(p + \lambda)$, integrating the inertial term by parts, so that $\ddot{q}$ disappears.

Code. Implement $\hat{M}_\theta = L_\theta L_\theta^\top + \epsilon I$; $\hat{C}_\theta$ from the Christoffel symbols of $\hat{M}_\theta$; $\nabla_q \hat{U}_\theta$ from a scalar potential head; $\hat{D}_\theta(\dot{q}) \succeq 0$ with nonnegative weights and monotone activations. Two ways to obtain $\partial \hat{M}_\theta / \partial q$ and $\nabla_q \hat{U}_\theta$ in MATLAB:

- `dlarray` with `dlfeval` and `dlgradient` (Deep Learning Toolbox). Note that `dlgradient` differentiates a *scalar*, so you loop over the $n(n + 1)/2$ independent entries of $\hat{M}_\theta$ — cheap for $n = 2$.
- For a one-hidden-layer network, write $\partial \hat{M}_\theta / \partial q$ analytically in half a page and code it directly. Faster, and it forces you to know what the derivative is.

Do the first, then check it against the second. Fit offline to $\Delta_{\text{el}} + \Delta_{\text{dis}}$ with the filtered cost (14) on Fourier-series excitation data from the sandbox, using `adamupdate` or `lsqnonlin`.

Exit test.

- $\|(\dot{\hat{M}}_\theta - 2\hat{C}_\theta) + (\dot{\hat{M}}_\theta - 2\hat{C}_\theta)^\top\| < 10^{-10}$ for random θ , q , $\dot{q}$, including untrained, randomly initialized θ . **This is the pass/fail test of whether you understood (N2).**
- $\lambda_{\min}(\hat{M}_\theta) > 0$ always.
- Training uses no measured acceleration anywhere — you must be able to point to the line of code where $\ddot{q}$ would have entered and show that it does not.

Deliverable. Note 6, plus the trained module and the reported δ_θ on the validation partition.

Phase 6 — First assembly and the simulated ablation

Goal. The full architecture (7) in simulation, and the ablation C1–C4 with the real numbers.

Tasks.

- Freeze the structure-generating layers; extract the typed regressor $\Sigma_\theta(x)$ with its four channels.
- Add the online law (17) with projection onto the admissible set $\mathcal{W}$.
- Add the multivariable STA with gains read from the measured $(\delta_\theta, L_\theta)$.
- Estimate L_θ numerically along closed-loop trajectories by finite differences on $\tilde{\Delta}$, and record it. This is the measured L_θ that will later sit next to the certified one in contribution C6.

- Run C1 (nominal PBC), C2 (PBC + learning), C3 (PBC + full-gain STA), C4 (complete), on the same trajectory, same noise seed, same h .

Exit test. C4 matches C3’s tracking error using strictly smaller k_1, k_2 , with lower $\int \|\tau\|^2$ and lower total variation. If it does not, diagnose the failure and write it down — that is also a valid outcome at this stage.

Deliverable. Table of the four configurations across all metrics of Section 5.9.

Phase 7 — Consolidation

- Merge Notes 1–6 into the state-of-the-art document.
- Write the derivation of (12), the three terms of the derivative certificate, as far as you can push it. You will not close it in this plan; that is second-term work. What you must produce now is a clear statement of which term is hard and why.
- Draft outline of the conference paper (target CDC or IFAC World Congress).

4. Parallel track: hardware

Platform commissioning always takes longer than planned, and the proposal already lists it as risk number one. Run this list in parallel with the theory phases, starting as soon as Phase 0 is running — not after Phase 7.

1. Identify the platform, read its manual, get encoder reading and torque command working at a fixed sampling period.
2. Measure the real noise floor of the encoder differentiation. This number sizes the dead zone in Section 5.4, and it is needed before Phase 3.
3. Implement filter (20) on hardware; compare against Levant’s differentiator [9].
4. Run the nominal PBC on hardware and record the experimental ρ_1 , next to the simulated one from Phase 2.
5. Torque budget check of Remark 5 — measure $\tau_{\max}$, and $\|\tau_{\text{pbc}}\|$ over the trajectory, so that the admissible Δ and $\nu_{\max}$ are known numbers, not symbols.

5. Read in the next term

These are deliberately postponed. Each becomes useful once the first-term material is in place, and each would cost a great deal of time if opened now.

- [16] — IDA-PBC and port-controlled Hamiltonian systems. Relevant if the second term extends the storage function beyond the EL form.
- [25] — adaptive PD for manipulators. The comparison point when you write the related-work section against classical adaptive control.
- [19] — two-stage neural observer for mechanical systems. Relevant when velocity estimation stops being a filter and becomes an observer.
- [28] — fuzzy identification with stable learning. The same Lyapunov argument on a different regressor; useful when you generalize Σ_θ .
- [17] — reinforcement learning for human–robot interaction. Not part of this thesis, but the natural continuation if the residual later becomes a learned policy.

- The general PINN/PDE literature beyond the single contrast reading of [18].
- Deep-learning architecture literature (transformers, residual networks, normalization tricks). The network here is small and shallow on purpose.
- Discrete-time sliding mode — open it only after Phase 4 is finished, when the $O(k^2h^2)$ result makes the question concrete.

6. Tooling

MATLAB

- Everything in MATLAB, one repository, one folder per phase. Toolboxes you will need: Symbolic Math (Phase 0), Deep Learning (Phase 5), Optimization (the offline fit), Simulink and a real-time target for the hardware track.
- **Fixed-step integration, always.** Write your own RK4 in Phase 0 and keep using it. Do not run the super-twisting algorithm under `ode45`: a variable-step solver shrinks its step near the discontinuity, hides chattering, and reports a sliding accuracy that no real implementation will reproduce. The whole point of experiment (B4) is that the accuracy depends on h , so h must be yours to set.
- Use `syms` and `matlabFunction` once, in Phase 0, to cross-check the hand-derived M , C , g , and then work with the generated functions — never with symbolic expressions — inside the integrator.
- Version control with git from day one. Write plain `.m` files with `%%` sections, not `.mlx` live scripts: live scripts do not diff, so you cannot see what changed between two versions of a controller.
- Tag the repository at every exit test. When something breaks during the assembly of Phase 6, you want to be able to return to the last configuration that passed.
- $\text{\LaTeX}$ for all notes, with the same `refs.bib` as the proposal, so the thesis assembles itself later.
- Real-time code (Simulink Desktop Real-Time, Speedgoat, or the platform’s own interface) only for the hardware track, and only once the nominal loop of Phase 2 runs in simulation.

Using Gemini or Claude

They will save real time on the parts that are not the thesis, and quietly damage the parts that are. The dividing line is whether the output can be checked by running it.

- **Good uses:** plotting and figure formatting, vectorizing a loop, MATLAB API questions, reading an error message, turning a script into functions, writing the data-logging and metric code, $\text{\LaTeX}$ tables and `tikz`, improving the English of a note you have already written.
- **Not for derivations.** The learning law (17), the trace cancellation in (16), the Moreno–Osorio conditions, the three terms of (12): this is where the models produce confident, well-formatted, wrong mathematics, and where you have no independent way to notice. Every “derive by hand” line in this plan means by hand.
- **Not for the numerical method.** Asked for a sliding-mode simulation, they will hand you `ode45` and a smoothed tanh in place of the sign function. Both destroy the experiment.
- **Nothing enters the repository until it passes a test you wrote.** Generated code goes in its own commit, so that it can be reverted in one step.
- **If you cannot explain a line at the board, delete it.** Not as a matter of principle: in Phase 5 the exit test is a 10^{-10} skew-symmetry check, and you will not be able to debug a

failure in code you do not understand.

7. Meeting protocol

- Each meeting, 45 minutes, with a fixed agenda: (i) the current exit-test number; (ii) one derivation at the board, chosen by the advisor; (iii) one figure; (iv) the single thing that blocked you.
- Alternating advisors from Phase 4 on: Dr. Yu on the learning and passivity side, Dr. Castaños on the sliding-mode side, joint meeting at the end of each phase.
- No slides. Notebook, board, and the current plot.

8. Mistakes to avoid

Write these on the first page of your notebook.

1. Using measured $\ddot{q}$ anywhere — in training, in the regressor, or in the controller. Remark 2 of the proposal exists precisely for this.
2. Adapting the structure-generating layers online. Only $\hat{W}$ moves; the rest is frozen.
3. Removing projection “because it never activates”. It is a term of the certificate (12), not a safeguard.
4. Comparing C1–C4 with different gains, different trajectories, different h or different noise seeds.
5. Reporting RMSE alone. Every tracking number is reported together with $\int \|\tau\|^2$, total variation of τ , and the gain magnitudes — otherwise the comparison is meaningless, since the whole claim of the thesis is that accuracy is bought with smaller gains.
6. Claiming that learning improves asymptotic accuracy under ideal STA. Remark 6 says the opposite, and reviewers know it.
7. Writing “physics-informed” or “passivity-based sliding mode” as if it were the contribution. Section 2 of the proposal says these labels are taken.

9. What this plan looks like

The proposal is correct and it is ambitious. You are not going to read it again until Phase 7. You are going to build it. Out of Phase 0 you get a 2-DOF arm you wrote yourself, with skew-symmetry verified to ten decimal places. Out of Phase 2 you get the baseline figure of your thesis. Out of Phase 4 you get the $O(h^2)$ figure, which is the strongest result in the proposal, obtained experimentally before anything is proved. Out of Phase 5 you get a network whose inertia matrix is positive definite before it is trained. Then, and only then, we assemble.

Two rules. Nothing is read that is not implemented immediately. Nothing is used that you cannot re-derive at the board.

Appendix: the reading list, in two groups

The proposal cites 30 references. You do not need 30. Five of them carry the argument of the thesis, and you should read those completely, in the order given. Everything else is opened when the phase that needs it arrives, usually for a single result, and put down again.

Group 1 — Must read, completely, in this order

#	Ref.	What it gives you	Phase
1	[7]	Manipulator dynamics, the structural properties (2)–(5), and PD / PBC regulation with complete Lyapunov proofs. Your base text.	0–1
2	[22]	The filtered error s and the nominal law (7) — the skeleton of the whole architecture.	2
3	[30]	The Lyapunov learning law, why the error converges to a ball, and why removing that ball needs an integral action.	3
4	[13]	The strict Lyapunov function of the super-twisting algorithm and the gain conditions your certificate has to close.	4
5	[11]	The structure-preserving parameterization: $\hat{M} = LL^\top$, Christoffel $\hat{C}$, $\nabla_q \hat{U}$.	5

These five, and only these five, you must be able to discuss at the board without notes.

Group 2 — The rest, one result each

Phase	Ref.	Open it for
0	[24]	Ch. 6–7: the Lagrangian derivation of M, C, g , if Kelly’s treatment is too compressed.
1	[15]	Ch. 1 and the EL chapter: the energy-shaping and damping-injection reading of the same controller.
2	[23]	Where the sliding surface $s = \dot{e} + \Lambda e$ comes from.
3	[10]	The canonical UUB proof with a two-layer net.
3	[20]	Why a linearly parameterized regressor has no local minima.
3	[27]	The identification-side version of the same learning law.
3	[6]	Ch. 8 only: e - and σ -modification, dead zone, projection.
4	[8]	The sliding-accuracy result $ s = O(k^2 h^2)$, which is contribution (B4).
4	[2]	Integral sliding manifolds and unmatched perturbations. Read it before your first meeting with Dr. Castaños.
4	[14]	The multivariable STA under the matrix gain $M^{-1}(q)$.
4	[4]	Variable-gain STA, the declared fallback.
5	[5]	Hamiltonian networks: the antecedent of DeLaN. One afternoon.
5	[3]	Lagrangian networks: the other antecedent. One afternoon.
5	[18]	Contrast only: physics as a penalty in the cost, rather than as structure in the architecture.
5	[21]	Spectral normalization used inside a stability certificate.
5	[1]	The acceleration-free regression for parameter identification.
5	[12]	The filtered form of the same idea.
6	[9]	The robust exact differentiator, as the alternative to filter (20).
6	[26]	Convergence time of the super-twisting algorithm.

The five references that appear in neither group — [16], [25], [19], [28], [17] — are the next-term list of Section 5. That accounts for all 30.

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